THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 12: **Quadrilaterals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The diagonals of a rectangle: 1

(a) are equal and bisect each other **(b)** are perpendicular
(c) are unequal **(d)** do not bisect each other

2. In a rhombus ABCD, $\angle A = 66^\circ$. The angle between the diagonal AC and the side AB is: 1

(a) 90° **(b)** 33° **(c)** 99° **(d)** 43°

3. The sides of a triangle are 20 cm, 22 cm and 16 cm. The perimeter of the triangle formed by joining the midpoints of its sides is: 1

(a) 31 cm **(b)** 87 cm **(c)** 29 cm **(d)** 43 cm

4. The diagonal of a square is 16 cm. Its area is: 1

(a) 144 cm^2 **(b)** 132 cm^2 **(c)** 160 cm^2 **(d)** 128 cm^2

5. The angles of a quadrilateral are in the ratio $2 : 6 : 4 : 3$. The largest angle is: 1

(a) 164° **(b)** 154° **(c)** 144° **(d)** 134°

6. In a parallelogram, two opposite angles are $(2x + 20)^\circ$ and $(x + 42)^\circ$. The value of x is: 1

(a) 22 **(b)** 21 **(c)** 23 **(d)** 24

7. Two adjacent angles of a parallelogram are in the ratio $1 : 2$. The smaller angle is: 1

(a) 70° **(b)** 80° **(c)** 50° **(d)** 60°

8. The diagonals of a parallelogram intersect at O. If $OA = (2x + 3)$ cm and $x = 7$, then the diagonal AC is: 1

(a) 38 cm **(b)** 34 cm **(c)** 36 cm **(d)** 51 cm

9. The sum of the interior angles of any quadrilateral is: 1

(a) 360° **(b)** 540° **(c)** 270° **(d)** 180°

10. In a rhombus ABCD, $\angle A = 56^\circ$. The angle between the diagonal AC and the side AB is: 1

(a) 90° **(b)** 38° **(c)** 84° **(d)** 28°

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. In $\triangle ABC$, $AB = 26$ cm, $BC = 48$ cm and $CA = 36$ cm. D, E, F are the midpoints of AB, BC and CA. Find the sides of $\triangle DEF$ and its perimeter. 2

12. In a parallelogram ABCD, $\angle A = 60°$. Find $\angle B$, $\angle C$ and $\angle D$. 2

13. The sides of a rectangle are 5 cm and 12 cm. Find the length of its diagonal and its perimeter. 2

14. In $\triangle ABC$, $AB = 24$ cm, $BC = 20$ cm and $CA = 42$ cm. D, E, F are the midpoints of AB, BC and CA. Find the sides of $\triangle DEF$ and its perimeter. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The opposite angles of a parallelogram are $(2x + 10)°$ and $(x + 27)°$. Find x and all four angles. 3

16. Two adjacent angles of a parallelogram are in the ratio 1 : 3. Find all four angles. 3

17. The diagonals of a parallelogram ABCD meet at O. $OA = (4x + 9)$ cm and $OC = (2x + 23)$ cm. Find x and the length of AC. 3

OR

The opposite angles of a parallelogram are $(4x + 7)°$ and $(x + 76)°$. Find x and all four angles. 3

18. Two adjacent angles of a parallelogram are in the ratio 1 : 2. Find all four angles. 3

OR

The diagonals of a parallelogram ABCD meet at O. $OA = (3x + 8)$ cm and $OC = (4x + 6)$ cm. Find x and the length of AC. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. In $\triangle ABC$, D is the midpoint of AB and the line through D parallel to BC meets AC at E. $BC = 24$ cm and $AB = 22$ cm. Find DE, AD and show that E is the midpoint of AC. 5

OR

The diagonals of a quadrilateral ABCD are AC = 12 cm and BD = 16 cm. P, Q, R, S are the midpoints of AB, BC, CD, DA. Show that PQRS is a parallelogram and find its perimeter. 5

20. In $\triangle ABC$, D is the midpoint of AB and the line through D parallel to BC meets AC at E. $BC = 30$ cm and $AB = 16$ cm. Find DE, AD and show that E is the midpoint of AC. 5

OR

The diagonals of a quadrilateral ABCD are AC = 18 cm and BD = 29 cm. P, Q, R, S are the midpoints of AB, BC, CD, DA. Show that PQRS is a parallelogram and find its perimeter. 5

END OF QUESTION PAPER — Total: 40 Marks

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